

EE6320 RF Integrated Circuits
Project: PA Design

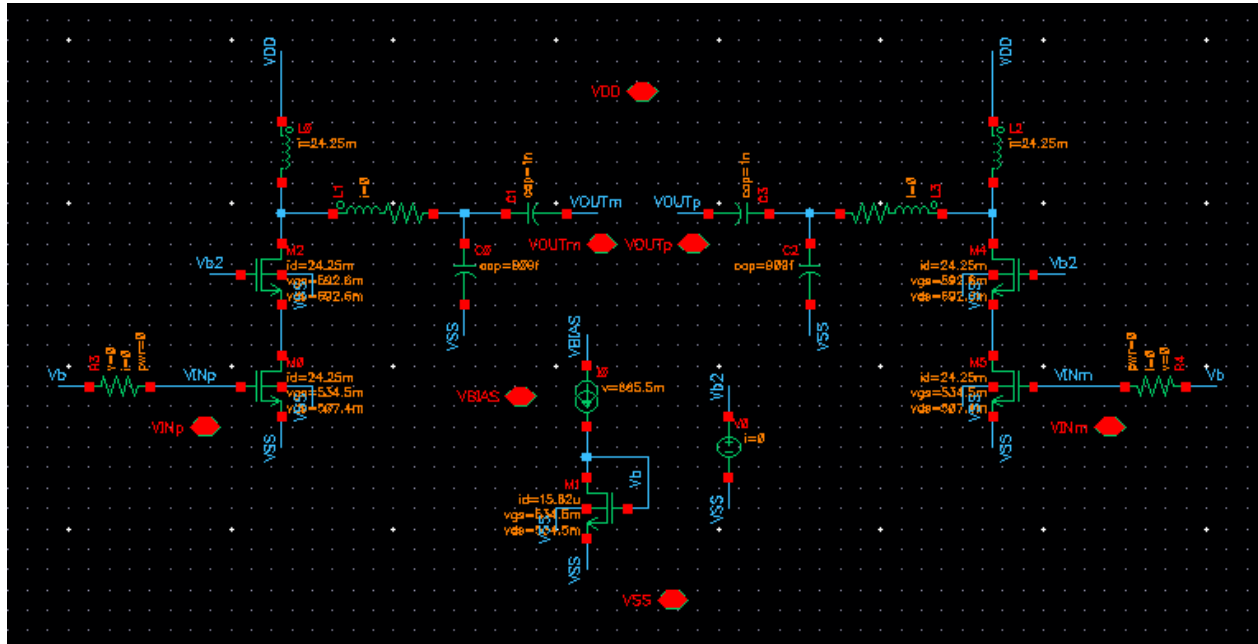
PA Performance Summary Table

Design Parameter	Design Metric	Performance	Specification
Output P1dB	$f_o = 3450 \text{ MHz}$	14.4483 dBm	$\geq +13 \text{ dBm}$
	$f_o = 3500 \text{ MHz}$	14.3793 dBm	$\geq +13 \text{ dBm}$
	$f_o = 3550 \text{ MHz}$	14.3175 dBm	$\geq +13 \text{ dBm}$
AM-PM Deviation (at P1dB)	$f_o = 3450 \text{ MHz}$	2.8609 degrees	$\leq 3 \text{ degrees}$
	$f_o = 3500 \text{ MHz}$	2.8474 degrees	$\leq 3 \text{ degrees}$
	$f_o = 3550 \text{ MHz}$	2.8176 degrees	$\leq 3 \text{ degrees}$
Voltage Gain (from Gate to Drain)	$f_o = 3450 \text{ MHz}$	33.66076	≥ 2
	$f_o = 3500 \text{ MHz}$	33.57486	≥ 2
	$f_o = 3550 \text{ MHz}$	33.49704	≥ 2
Power (at 3.5 GHz)	PA Average Consumption (excluding bias)	80.1732 mW	Minimize

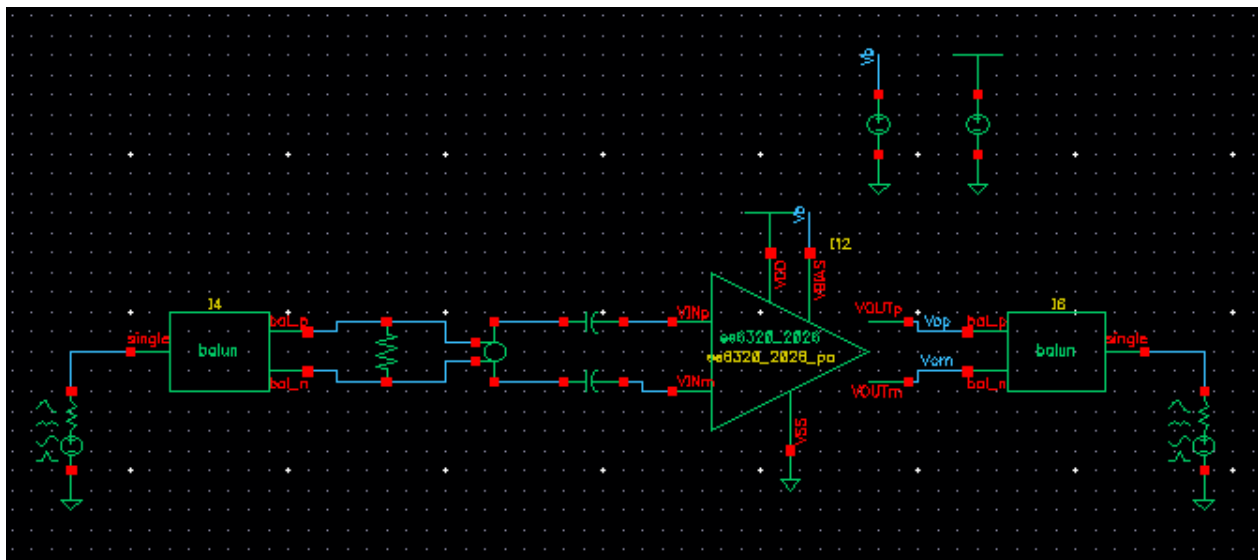
	Bias Circuit Consumption	18.75 μW	Minimize
Other	Sum of all Capacitances (including ac coupling)	4.001818 nF	-
	Inductance Used	2.2736 nF	-
	Device Width	1600 μm	-
	Simulator Used	Spectre	-

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PA Schematic



PA Testbench

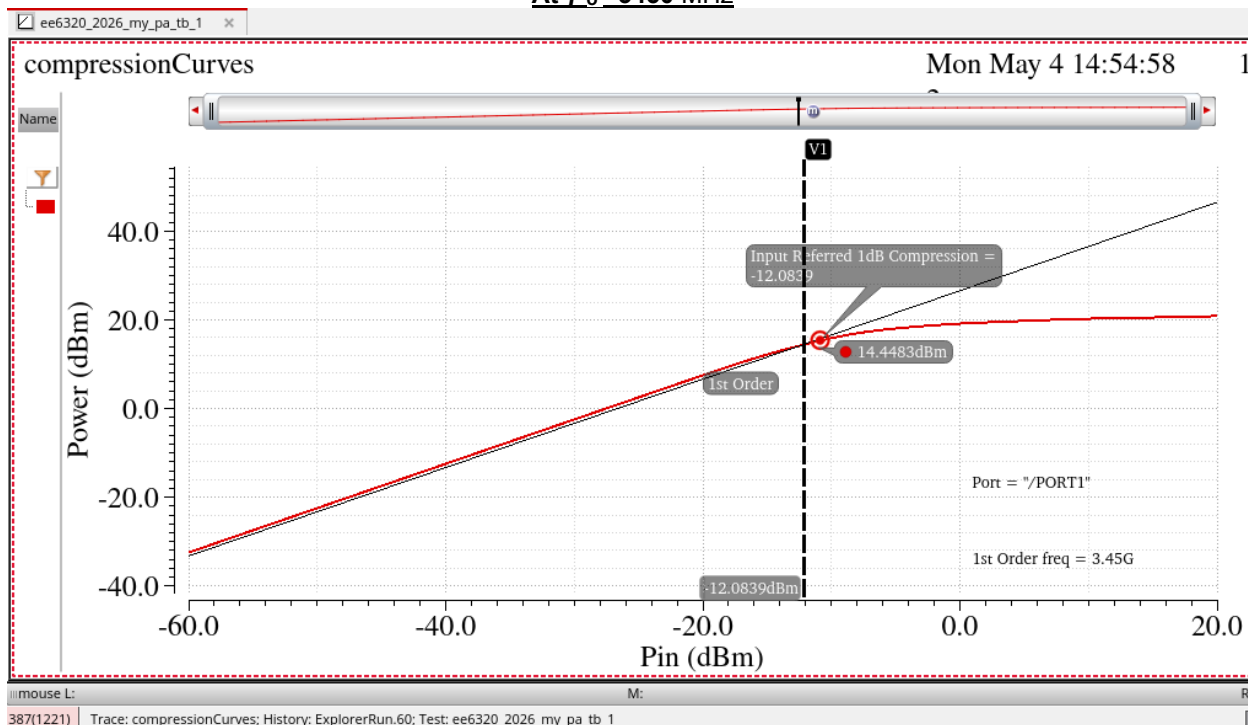


Component Values Table

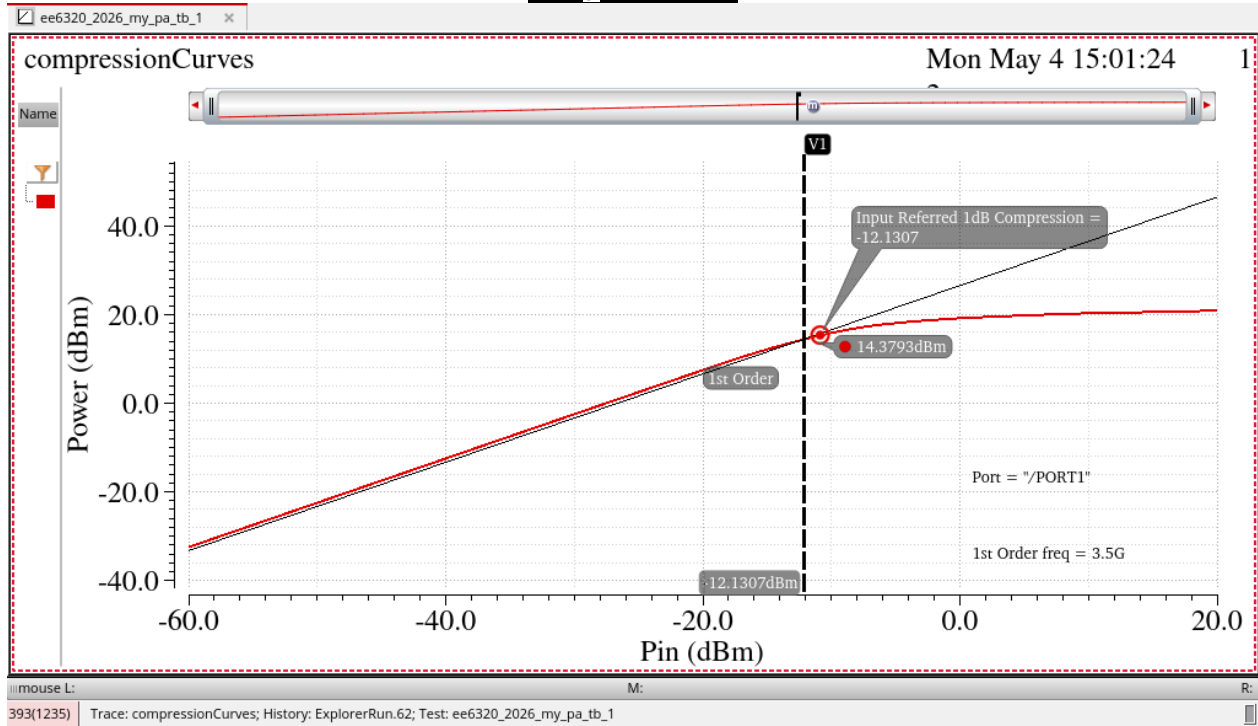
(W, L) MOS Amplifier = (1600 μm , 0.12 μm)	(L, C) Matching network = (1.1368 nH, 0.909 pF)
(W, L) Current Mirror MOS = (1 μm , 0.12 μm)	Bias Current = 15.625 μA
R_bias = 100 k Ω , C_coupling = 1 nF	V_DD = 1.2 V, R_load = 50 Ω

Output 1dB Compression Point Plots

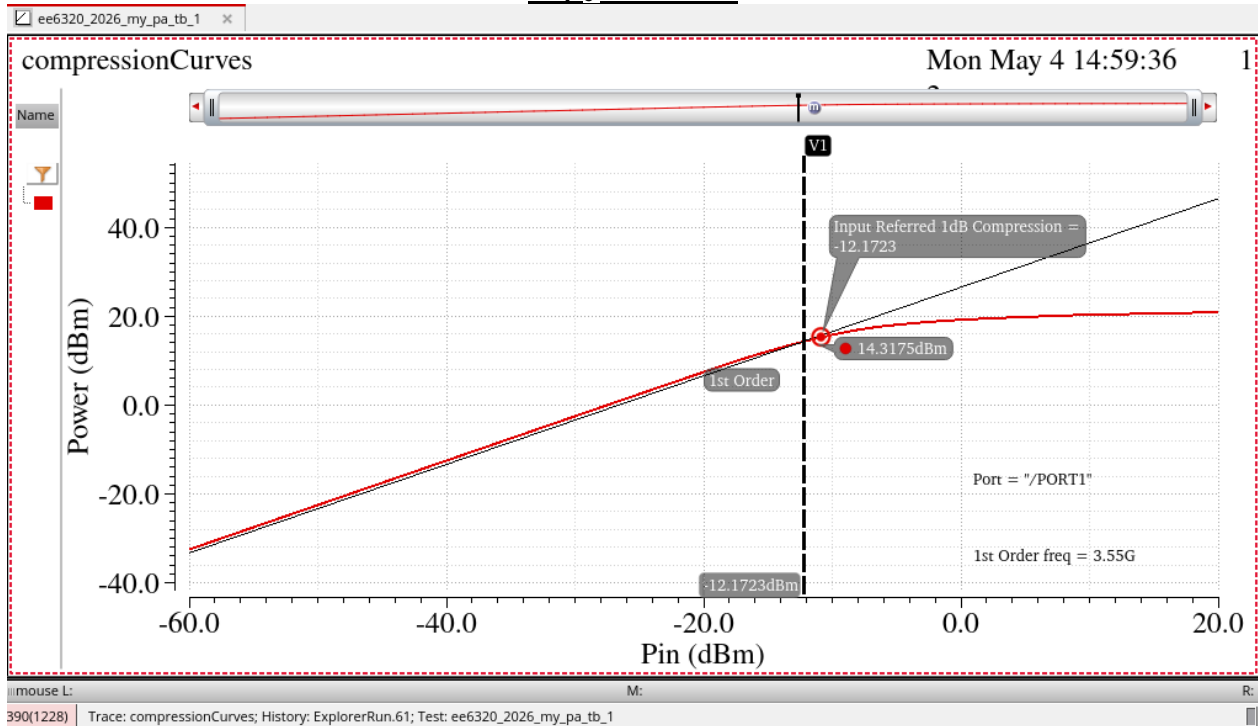
At $f_o = 3450$ MHz



At $f_o = 3500$ MHz

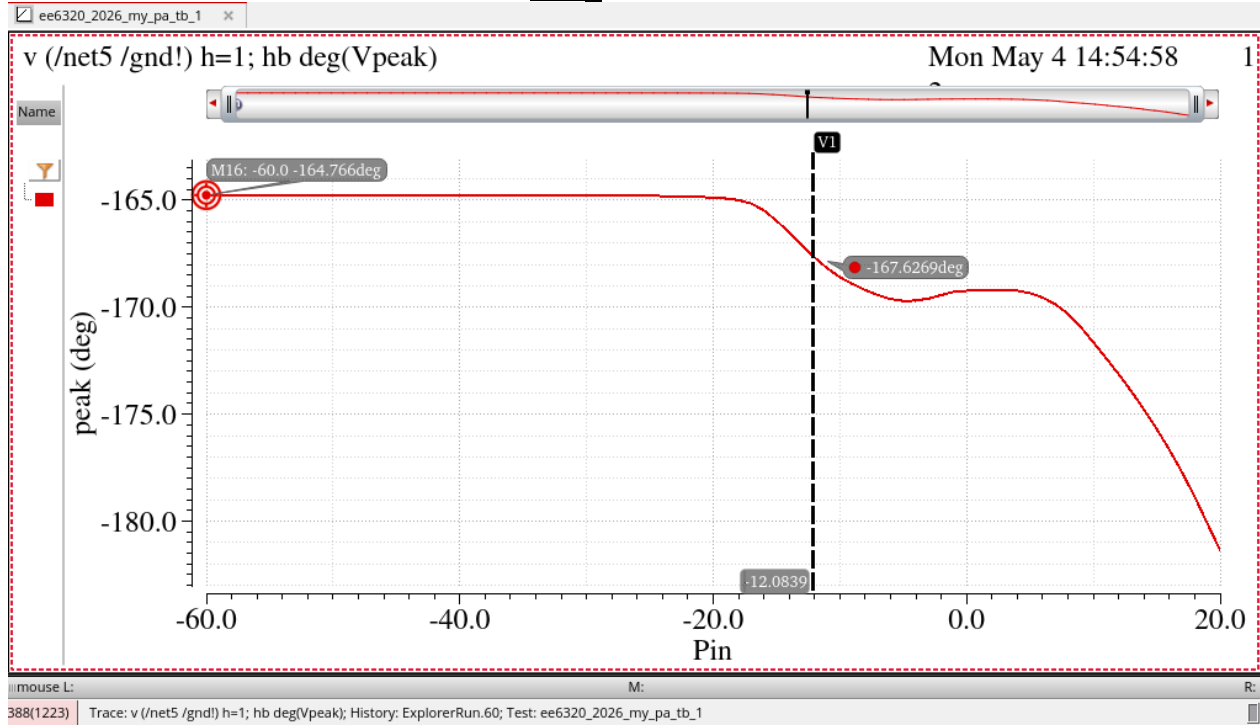


At $f_o = 3550$ MHz

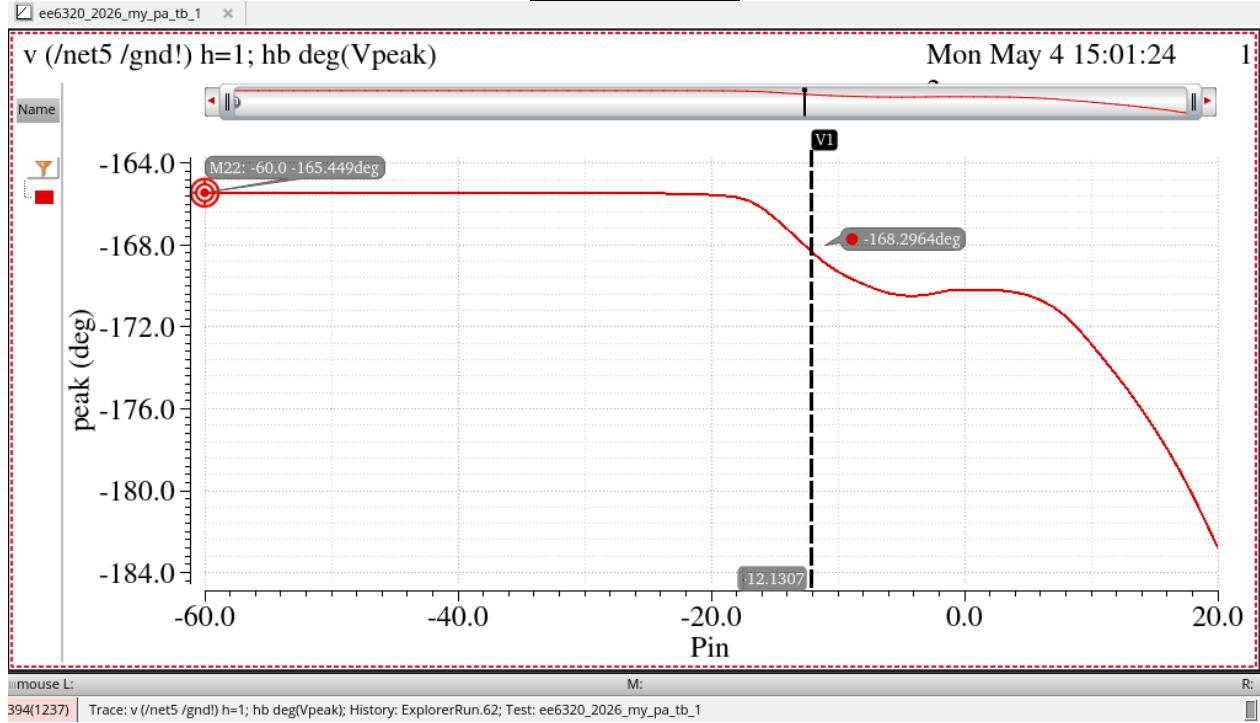


AM-PM Deviation Plots

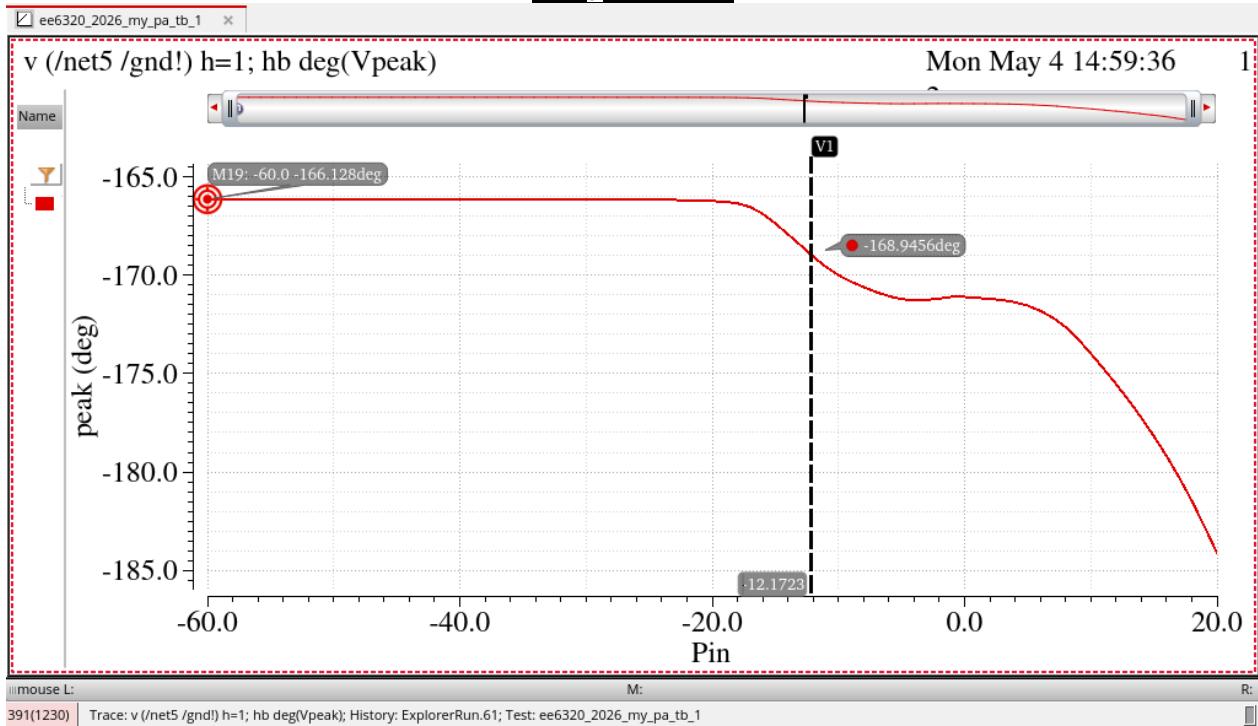
At $f_o = 3450$ MHz



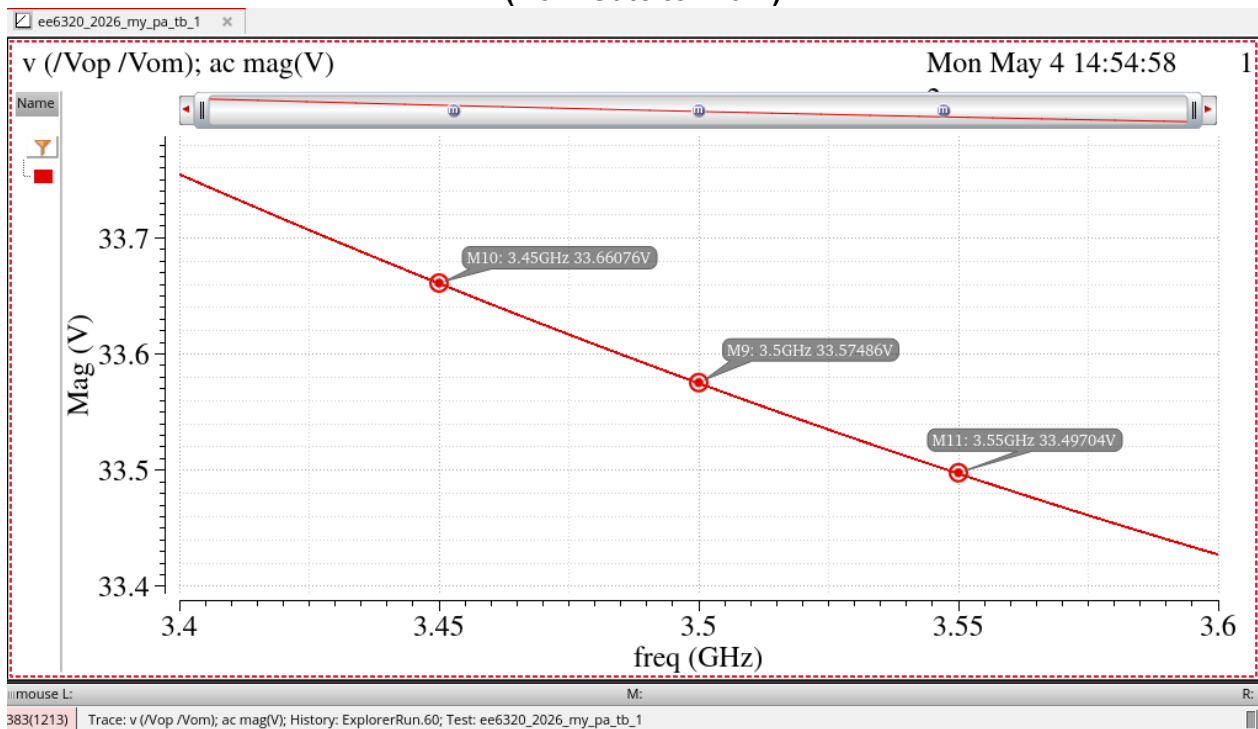
At $f_o = 3500$ MHz



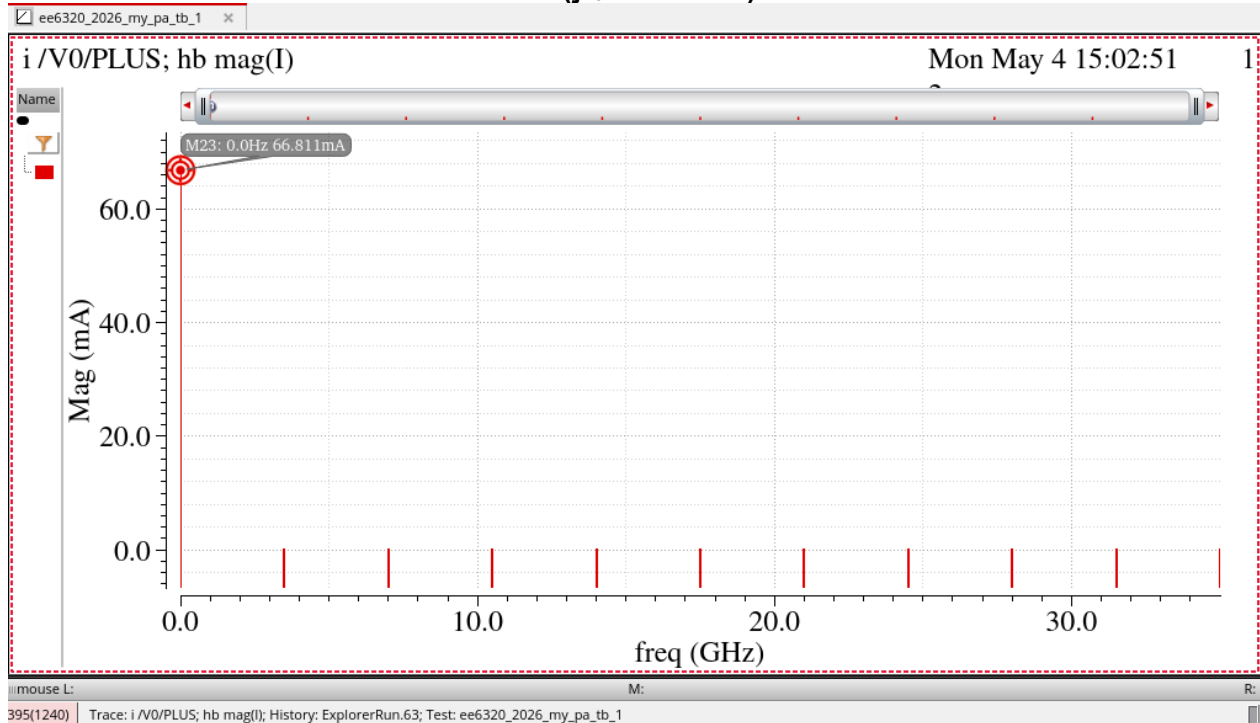
At $f_o = 3550$ MHz



PA Voltage Gain Plot
(from Gate to Drain)



Current through VDD Plot ($f_o = 3500 \text{ MHz}$)



Design Procedure

- Based on the hand calculations for the allowable V_a , we find a suitable R_{load} . We need to use a matching network to achieve this R_{load} . To find the best-suited matching network, we add a port across the matching network and then see which design gives the best $\text{Re}(Z_{11})$ and $\text{IM}(Z_{11})$ for the desired value of R_{load} .
- Then, we simulate the gain, which will most likely pass the specification.
- We then simulate the P1dB and AM-PM curves.
- The width of the MOSFET should be swept to find the optimum point where both the specifications are met.
- If no such point is found, the bias current can be increased and the width swept again till the specifications are met.
- The cascode can be omitted in case the triode limits affect the non-linearity.
- Then, we can optimize for the power consumption (for a fixed bias current) by changing the width finely or by changing the bias voltage of the cascode, looking for the point of minimum current consumption.
- All MOSFETs need to be maintained in the saturation region throughout all of the above steps.

Path to Project Files

Schematic: /home/ep22b020/cadence/EE6320_2026/ee6320_2026/ee6320_2026_pa

Testbench: /home/ep22b020/cadence/EE6320_2026/ee6320_2026/my_pa_tb